

CS6440: Introduction to Health Informatics Spr24

Research for SkinWise Project

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Abstract—We conducted a literature review and explored potential resources for our project, *SkinWise*, a web-based diagnostic tool aimed at empowering healthcare providers in the early detection and effective management of skin cancer. Below, we outline our findings from the literature review and identify tools and resources that we can use for our project.

LITERATURE REVIEW

The Significance of Skin Cancer Detection in The Health Domain

Skin cancer is a major public health issue, with its incidence increasing globally. Among various types, melanoma is particularly notable for its severity and high mortality risk if not identified early (Khazaei et al., 2019). The standard approach to skin cancer diagnosis has traditionally relied on visual inspection and dermoscopic analysis of skin lesions (Ruffano et al., 2018). The accuracy of such diagnoses often varies, largely due to the subjective nature of the assessment, which is dependent on the individual healthcare professional's experience and expertise. This variability can lead to diagnostic inaccuracies.

The challenge of diagnosing skin cancer lies in the subtlety of its manifestations and the requirement for specialized knowledge to interpret dermoscopic images accurately (Loescher et al., 2013). Given these complexities, the proposed project, *SkinWise*, aims to leverage advancements in artificial intelligence (AI) and a comprehensive dataset from the Skin Cancer ISIC Dataset available at <https://www.isic-archive.com/>. By integrating these resources with Fast Healthcare Interoperability Resources (FHIR), *SkinWise* seeks to augment the current diagnostic processes and improve patient management strategies (Ruffano et al., 2018). This initiative offers a promising solution to the challenges of skin cancer diagnosis by enhancing the objectivity, accuracy, and efficiency of the process (Patel et al., 2023).

The deployment of an automated skin cancer detection system (Dildar et al., 2021) is critical in addressing the global disparities in access to specialized dermatological care. Conventional diagnostic methods' subjective nature makes it

unreliable, emphasizing the need for a standardized and objective diagnostic tool (Naqvi et al., 2023).

The Importance of Interoperability in The Health Informatics

In the realm of healthcare, interoperability stands as a cornerstone for efficient and effective patient care (Sreenivasan and Chacko, 2021). The ability of diverse systems and technologies to seamlessly communicate and exchange data is paramount in ensuring comprehensive and holistic treatment for patients. In this context, we want to find a transformative solution to the challenges of skin cancer diagnosis. By prioritizing interoperability, we hope our web tool could set itself apart as a platform that not only analyzes images of skin lesions but also seamlessly integrates with various electronic health record (EHR) systems.

The incorporation of Fast Healthcare Interoperability Resources (FHIR) standards within our web tool will signify a commitment to bridging the gaps between disparate healthcare systems. FHIR standards facilitate the standardized exchange of healthcare information, enabling healthcare providers to access critical patient data efficiently. With certain degree adherence to FHIR standards, healthcare professionals can seamlessly upload patient images and receive diagnostic suggestions directly within their existing EHR workflows.

This emphasis on interoperability not only streamlines the diagnostic process but also fosters collaboration among healthcare providers. By breaking down silos and enabling data exchange, the web tool is expected to empower healthcare teams to make informed decisions and deliver optimal care to patients. By prioritizing compatibility with existing electronic health record systems through the use of FHIR standards, we seek to enhance collaboration among healthcare providers and streamline the diagnostic process. This approach reflects a pragmatic and patient-centered approach to leveraging technology for improved healthcare delivery.

In summary, we expect that SkinWise can address these challenges by providing a web-based diagnostic platform that employs deep learning algorithms trained on a vast dataset. This approach aims to standardize skin cancer diagnosis. By integrating with FHIR standards, SkinWise also ensures seamless interoperability within the healthcare system.

RESOURCES

Useful Tools/Technology

A list of potential tools and technologies:

- **Python:** A versatile programming language selected for its extensive libraries and community support in backend development, data analysis, and machine learning classification model development.
- **PyTorch/TensorFlow/Keras:** These libraries are for developing and training machine learning models. PyTorch and TensorFlow provide comprehensive ecosystems for deep learning, while Keras simplifies neural network creation.
- **Flask:** A lightweight Python web application framework, chosen for its simplicity and flexibility in developing backend services.
- **HTML/CSS/JavaScript:** Core technologies for building the frontend, ensuring the application is responsive and interactive.
- **PostgreSQL:** An open-source relational database system, selected for its reliability and performance in managing application data.
- **Heroku:** A cloud platform as a service (PaaS) for easy deployment and scaling of web applications.
- **FHIR (Fast Healthcare Interoperability Resources) standards:** Adopted to ensure interoperability and secure communication of health information within the application.

Datasets and Data Sources

Potential datasets and data sources:

- **FHIR (Fast Healthcare Interoperability Resources):** To ensure interoperability in healthcare applications.
- **Skin Cancer ISIC dataset from Kaggle:** A comprehensive dataset of skin cancer images used for training the machine learning model.
- **Synthetic test data generated from the Skin Cancer ISIC dataset:** For testing purposes, ensuring the model's robustness and accuracy.

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